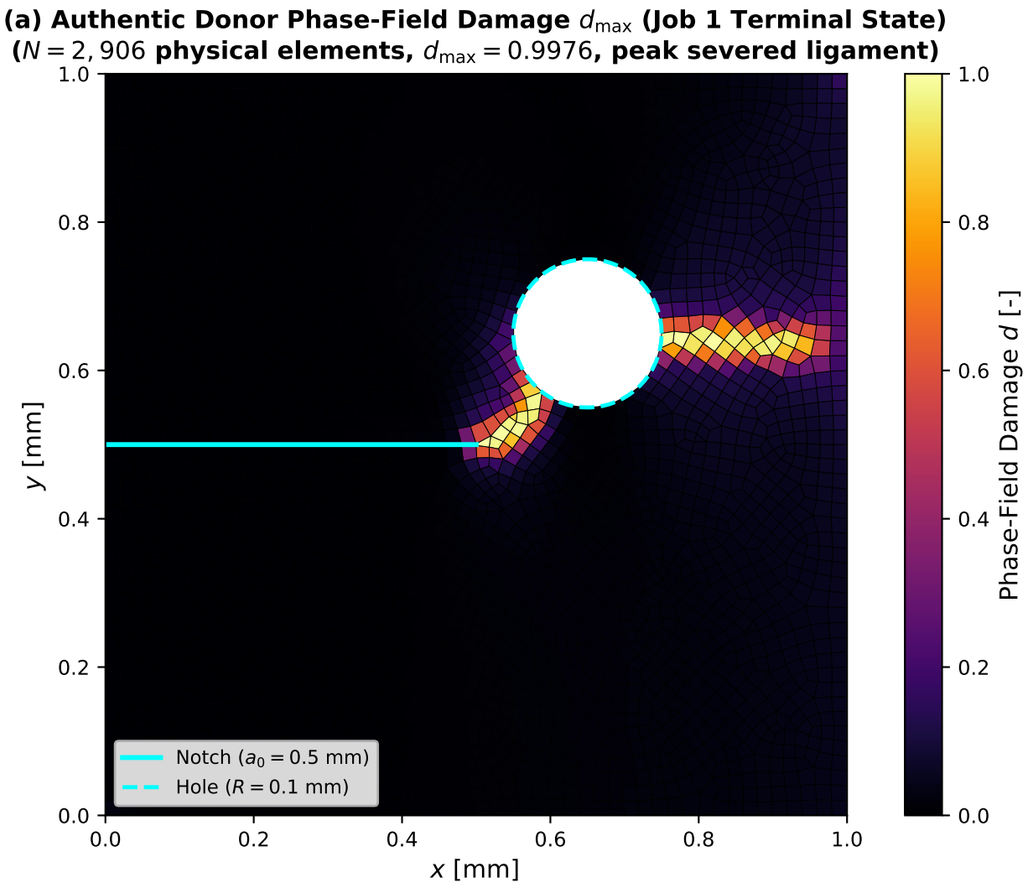
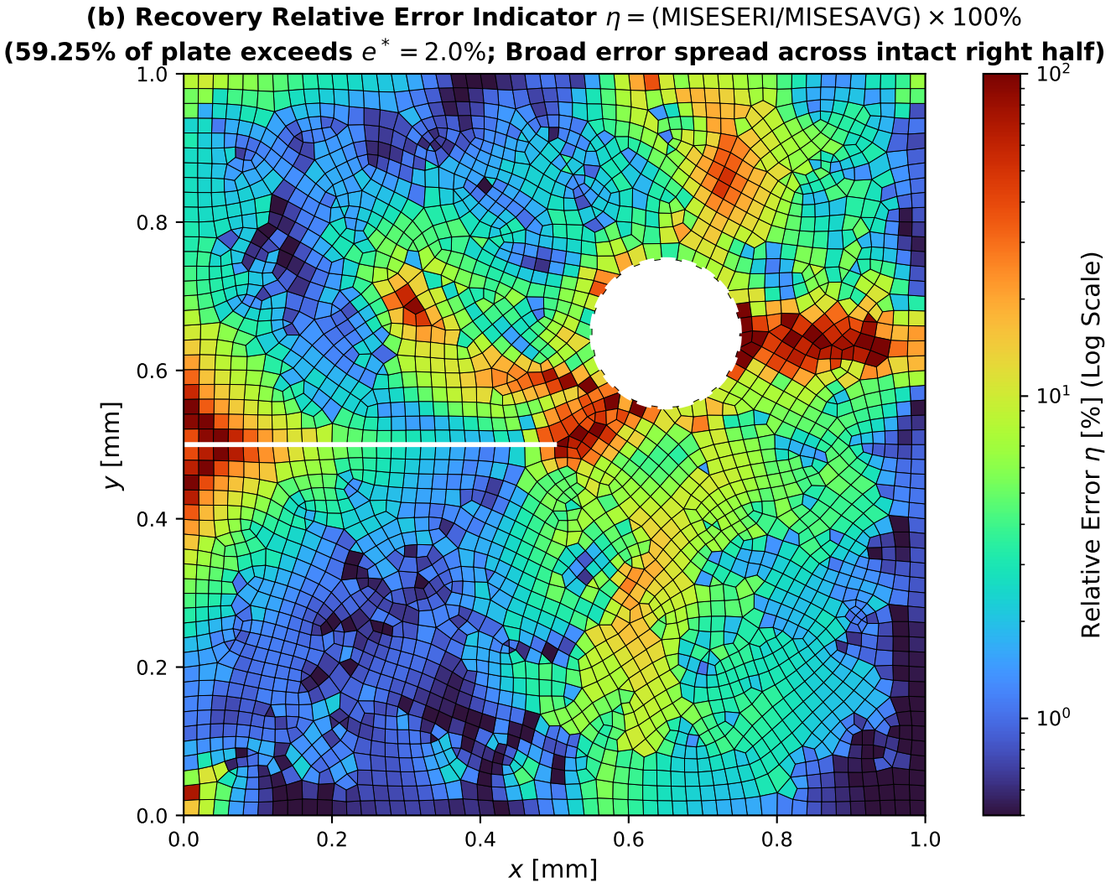
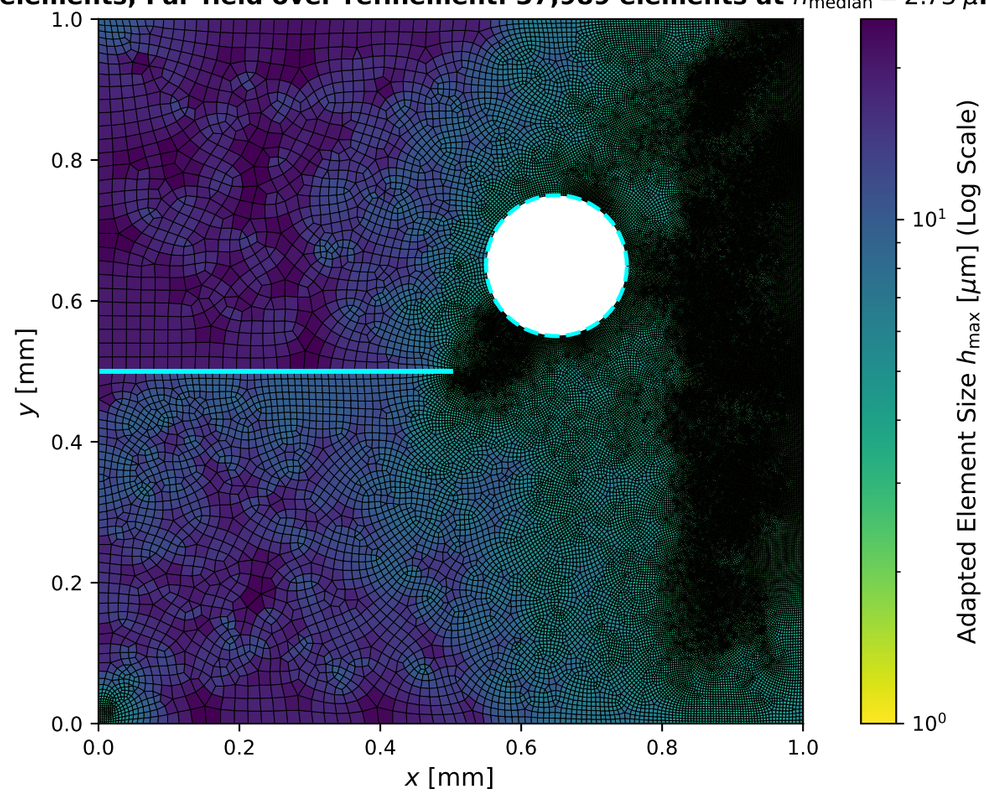
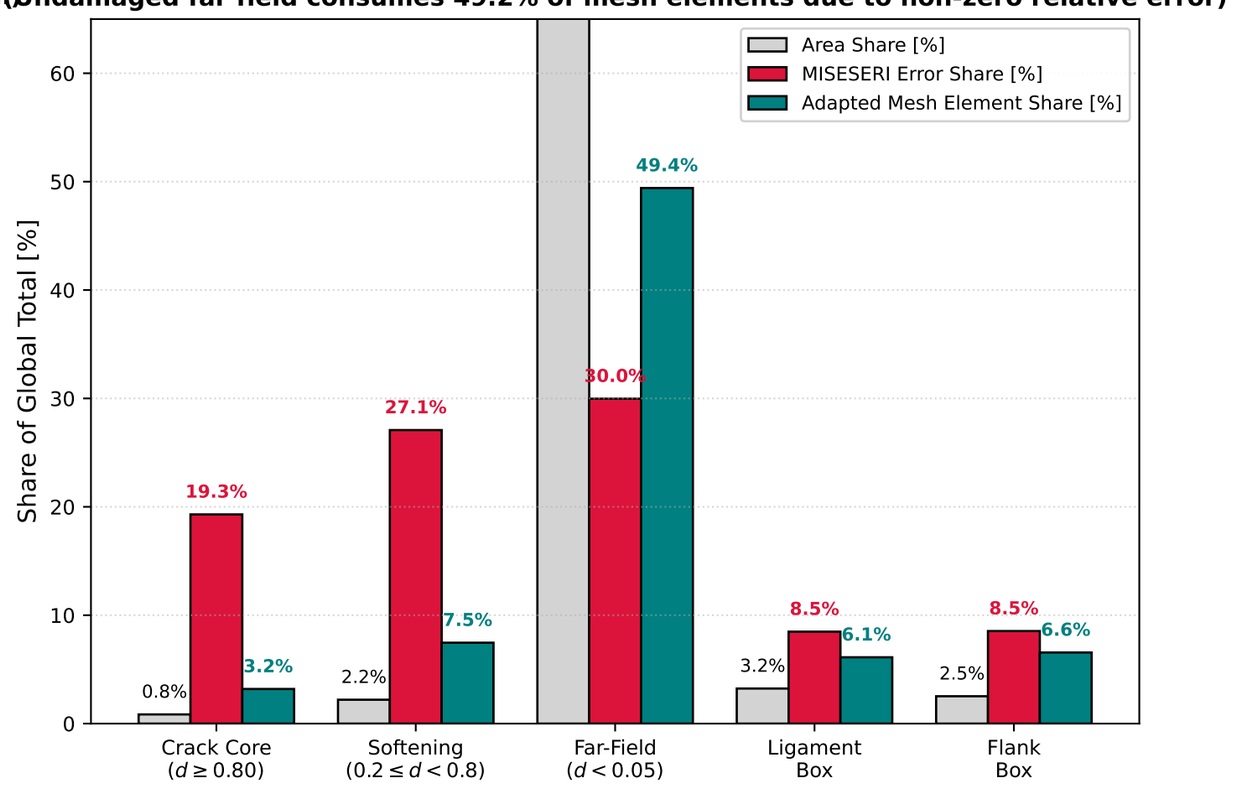

(a) Authentic Donor Phase-Field Damage d_{max} (Job 1 Terminal State)
($N = 2,906$ physical elements, $d_{max} = 0.9976$, peak severed ligament)
y [mm]
x [mm]
Notch ($a_0 = 0.5$ mm)
Hole ($R = 0.1$ mm)
Phase-Field Damage d [-]


(b) Recovery Relative Error Indicator $\eta = (\text{MISESERI}/\text{MISESAVG}) \times 100\%$
(59.25% of plate exceeds $e^* = 2.0\%$; Broad error spread across intact right half)
y [mm]
x [mm]
Relative Error η [%] (Log Scale)


(c) Resulting Adapted Mesh Resolution h_{max} [μm]
(77,211 elements; Far-field over-refinement: 37,989 elements at $h_{median} = 2.73$ μm)
y [mm]
x [mm]
Adapted Element Size h_{max} [μm] (Log Scale)


(d) Quantitative Causal Mismatch: Far-Field Cost vs Core Under-Allocation
(Undamaged far field consumes 49.2% of mesh elements due to non-zero relative error)
Area Share [%]
MISESERI Error Share [%]
Adapted Mesh Element Share [%]
Share of Global Total [%]
0.8%
19.3%
3.2%
2.2%
27.1%
7.5%
30.0%
49.4%
3.2%
8.5%
6.1%
2.5%
8.5%
6.6%
Crack Core ($d \geq 0.80$)
Softening ($0.2 \leq d < 0.8$)
Far-Field ($d < 0.05$)
Ligament Box
Flank Box